WHITEPAPER

Overcoming SOC Inaccuracies in LFP Batteries:

How to Increase Revenue and Simplify Operations

ACCURE
Battery Intelligence

EXECUTIVE SUMMARY

In recent years, Lithium Iron Phosphate (LFP) batteries have become the preferred choice for most large-scale storage systems, overtaking Nickel Manganese Cobalt (NMC) batteries due to their cost-effectiveness, improved safety, and supply chain advantages.

However, LFP batteries come with a significant drawback—they make it markedly difficult to calculate an accurate State of Charge (SOC). As a result, SOC errors exceeding 20% are now commonplace, leading to industry-wide performance issues such as non-delivery or missed revenue opportunities during peak demand periods.

For energy storage operators, precise and reliable SOC estimation can make the difference between hitting revenue targets or taking a financial loss. This paper examines the root causes of SOC inaccuracies, analyzes how various estimation methods contribute to these errors, and highlights the role of cloud-based predictive battery analytics in addressing these challenges and boosting SOC accuracy, and therefore – revenue.

THE CONSEQUENCES OF SOC INACCURACIES

Inaccurate SOC estimations pose big financial risks, especially when batteries are integrated into energy markets and grid services. These inaccuracies often lead to two primary financial consequences:

1. **Trading on incorrect energy and power volumes**
2. **Incurring penalties for non-compliance with market conditions.**

Trading Implications

When the SOC is overestimated or underestimated, trading decisions may result in selling either too much or too little energy or power. This misalignment reduces asset utilization, often causing immediate losses in revenue due to missed opportunities in periods of high demand or underperformance in energy trading.

Market Penalties

In energy and grid services markets, batteries are typically contracted based on their ability to deliver a specific power output for a set duration. An inaccurate SOC can cause a battery to fail to meet contractual obligations, leading to financial penalties.

In severe cases, operators could be excluded from these markets altogether, jeopardizing their long-term financial viability.

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CAUSES OF SOC INACCURACIES

The causes of SOC inaccuracies lie in how SOC is estimated. Generally, there are two main approaches:

1. **Coulomb counting (Ah-counting)**
2. **Voltage method**

Ideally, these two approaches are combined to improve accuracy. The specific way these methods are combined, though, depends on the sophistication of the Battery Management System (BMS).

TYPES OF BMS AND THEIR SOC ESTIMATION METHODS

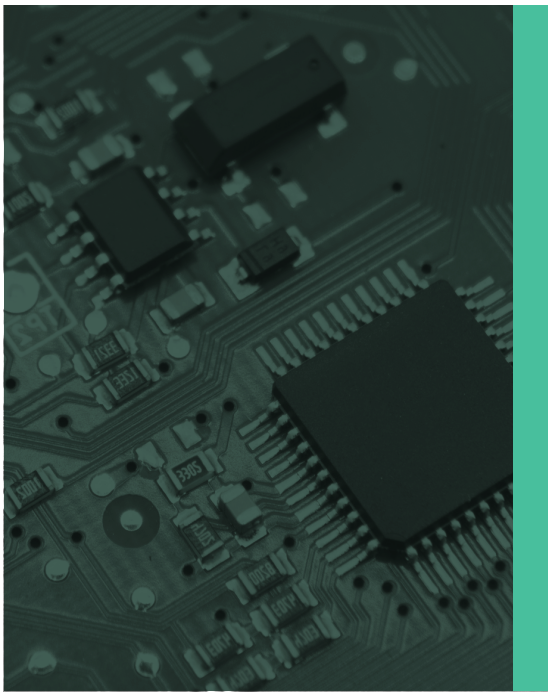
Battery Management Systems (BMS) differ in their methods for estimating SOC, each with varying levels of sophistication and accuracy. The way these systems integrate the voltage method and coulomb counting defines their effectiveness across various applications.

Rudimentary BMS

Rudimentary battery management systems (BMS) calculate SOC solely based on voltage method. As a result, their accuracy can be relatively poor. They're inexpensive, as they don't require a current sensor, and are often used in smaller applications like power tools. This type of BMS isn't suitable for larger battery storage systems.

Standard BMS

A step up from rudimentary BMS, standard BMS calculate SOC using a combination of voltage method and coulomb counting. Although used across almost all battery applications, including large BESS and automative battery systems, this type of BMS can be prone to SOC errors. This is because standard BMS mainly relies on coulomb counting to calculate SOC, which accumulates inaccuracies over time. Voltage method is used to recalibrate the SOC calculation, but it's only triggered when certain conditions are met, such as a full charge or relaxation phase, which can be infrequent in certain battery applications.



Advanced BMS

Lastly, there is the advanced BMS that apply mathematical techniques from control theory, such as Kalman Filters, to estimate the SOC. Non-linear versions of Kalman Filters, particularly the Extended Kalman Filter (EKF), are widely used, with some BMS even implementing an Unscented Kalman Filter for greater accuracy. These techniques rely on simplified electrical models of the battery, allowing them to be classified as "model-based" diagnostics. However, beneath the sophisticated math lies the age-old combination of coulomb counting and the voltage method.

Here's the takeaway: regardless of the sophistication level of the BMS, it still relies on coulomb counting and the voltage method, to some degree, to estimate SOC.

Although more advanced BMS try to sqeeze the best out of the two methods, they still face the exact same challenges as every other type of system. No BMS can maintain high accuracy in all conditions; there are always scenarios that lead to significant inaccuracies. This is especially true for applications where full (dis)charge scenarios are rare. Additionally, batteries with LFP cathodes pose unique diagnostic challenges, even for advanced BMS (see section on the voltage method below).

So, where exactly are these methods flawed? We'll explore below.

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CHALLENGES WITH COULOMB COUNTING

Capacity needs to be estimated continuously, but not all BMS can estimate it accurately.

Coulomb counting, or "Ah-counting", simply sums up the amount of charge (in ampere hours, Ah) charged or discharged into a battery. It then divides this sum by the battery's capacity, resulting in the SOC. A BMS calculates SOC iteratively for each time step using the following formula:

$$SOC_{k+1} = SOC_k + \frac{\Delta Ah}{C} \cdot 100\%$$

$$\Delta Ah = I \cdot \Delta t$$

Example: Assume a 100 Ah (C) battery is currently at 50% (SOC_k). By charging it with 10 A (I) for 2 h (Δt), 20 Ah (ΔAh) have been added to the battery, causing the SOC to increase by 20% to 70% (SOC_{k+1}).

At first glance, this method seems sound as a bell. Assuming that the initial SOC is known, then every subsequent SOC can be easily calculated. So, why does this method become inaccurate? The main issues stem from capacity estimation and inaccuracies in the measured current rate.

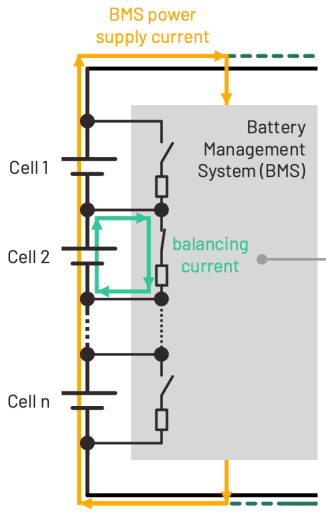
1. Capacity (C) Estimation:

The capacity value needed to calculate SOC is the battery's current usable capacity, which changes over time as the battery ages. Therefore, this capacity needs to be estimated continuously, but not all BMS can estimate it accurately; some rely on a rough guess or nominal capacity. **Any error in the capacity value directly causes an SOC estimation error.**

For example, consider a battery at its end-of-life criterion, with 80% residual capacity. If a BMS uses the nominal capacity to estimate SOC, a fully discharged battery (correctly at 0% SOC) would incorrectly be reported at 20% — an unacceptable error in most applications.

2. Measured Current Rate (*I*) Inaccuracies.

The second factor affecting SOC accuracy is the measured current rate. Generally, there are three common reasons for inaccurate current measurements. *Figure 1* illustrates this issue:



Not measuring the current drawn to power the BMS itself.

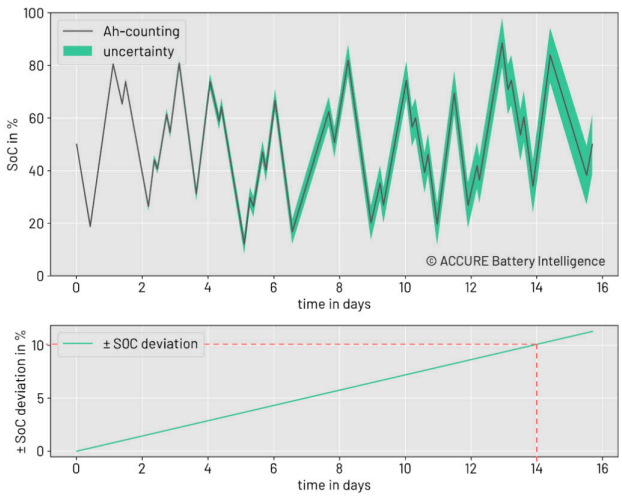
The **orange current flow** represents the current drawn from the battery to power the BMS. If the BMS powers itself from the cells it monitors, this energy drain is often not detected by the current sensor.

Not measuring the current drawn to balance the cells

Cell balancing is another energy drain, shown by the **light green current flow** in *Figure 1*. During balancing, the BMS connects resistors to individual cells to equalize their SOCs. While the homogenization is beneficial, this balancing current is typically not picked up by the current sensor, even though it reduces the battery's charge.

Figure 1a

Different effects that cause inaccurate current measurements



Current sensor having an offset error (or a nonlinearity)

The third source of error is the **current sensor itself**. No current sensor is 100% accurate; all have some degree of error. Although some measurement errors average themselves out over time, offset errors do not. Instead, they accumulate and contribute to an ongoing SOC error.

Figure 1b

Impact on the SOC calculation via Ah-counting.

Figure 1b demonstrates the concept of accumulating errors. Specifically, it illustrates how an SOC calculated via coulomb counting, changes over time. An offset error, whether positive or negative, causes SOC estimates to diverge from true SOC.

Example: If the same 100 Ah battery from the example above has a small offset error of 30 mA, after two weeks, the SOC estimation would be off by over ±10 Ah , leading to a ±10% SOC error.

Over a month or two, this drift will make the SOC estimation entirely useless. This is a huge problem.

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CHALLENGES WITH VOLTAGE METHOD

Using the voltage method to estimate SOC involves two steps. First, the open circuit voltage (OCV) is estimated of the battery at rest. Then, the estimated OCV value is converted into an SOC estimate using the battery's pre-determined OCV vs. SOC curve, unique to the battery's chemistry.

This curve is obtained through extensive testing and represents the relationship between the battery's voltage at rest and its charge level. As shown in *Figure 2*, the estimated OCV value is mapped onto this curve to determine the corresponding SOC, often by interpolating for values between known points.

While coulomb counting can lead to SOC estimates drifting over time, the voltage method can provide recalibration points to get the SOC estimation back on track. Because of this, the voltage method and coulomb counting complement each other well – in principle. In reality, though, this combination is prone to inaccuracies.

On the surface, converting an OCV value to an SOC value seems straightforward, and for some battery technologies, it is. However, this method faces two challenges, especially for LFP batteries:

1. The Hysteresis Effect

LFP batteries exhibit a "hysteresis effect," where the OCV varies depending on the recent charge or discharge history. *Figure 2* simplifies this with two different OCV curves: one for a battery primarily charged recently (**light green curve**) and one for a battery primarily discharged recently (**dark green curve**). The true OCV will lie somewhere between these two curves, depending on the battery's recent history.

For example, if the SOC was being estimated from the OCV curves in *Figure 2* without accounting for this effect, it'd be estimated at 27% for the battery primarily charged recently, and 62% for the battery primarily discharged recently, for the same OCV value of 3.3V. This difference in SOC estimation is more than 30 percentage points, which is essentially the maximum SOC error that can occur.

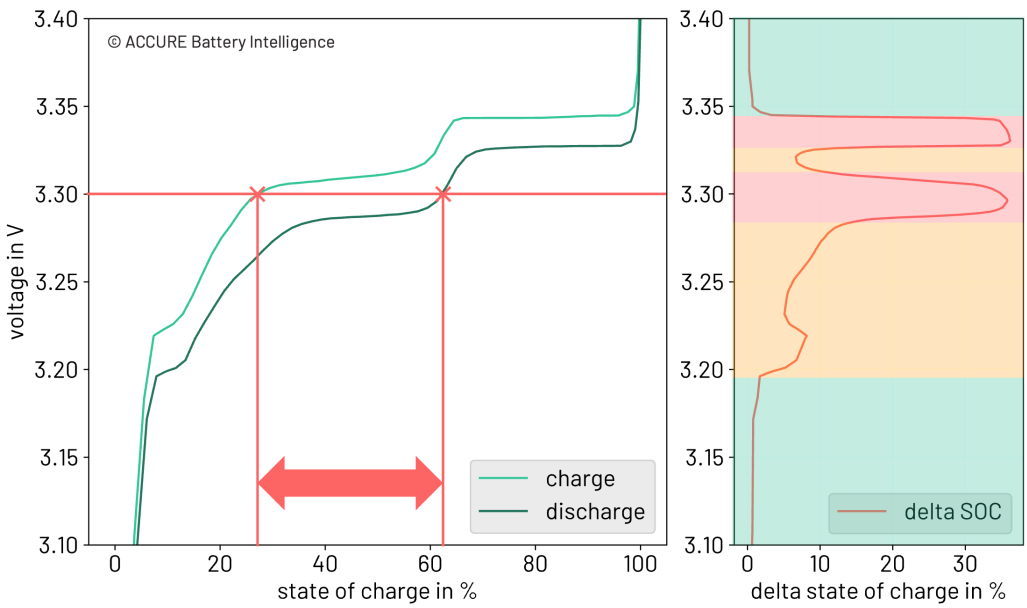
2. The Flatness of the OCV Curve

In flat sections, even small inaccuracies in estimating OCV can lead to large SOC errors. *Figure 2* shows several flat sections in the OCV curve of an LFP/graphite cell, which span a notable amount of the overall SOC range on the x-axis. Other lithium-ion technologies, such as those with lithium titanate oxide (LTO) anodes, face similar challenges.

To make matters more difficult, most battery applications including BESS operate almost constantly in closed circuit conditions (when the battery is in use), making estimation of the open circuit voltage (OCV) complex, requiring sophisticated modeling.

Figure 2

OCV curves of an LFP/graphite battery cell (left). SOC difference between the two OCV curves (right).



This leaves two options: The BMS can wait for occasional rest phases to recalibrate SOC, or it can try to estimate OCV under closed circuit conditions. In the first scenario, recalibration of the SOC is left to chance. This is why advanced BMS tend to choose the latter, using electrical models to simulate internal battery processes and estimate the hidden OCV behind all the other overvoltages. This is a complex task, as models are limited by computational power and must be updated continuously to account for battery aging.

Another layer of complexity is that the OCV curve changes with age, depending on the aging mechanisms affecting the battery. This topic is extensive and covered in more detail in our recent blog article.

Cloud-based predictive battery analytics offer a new perspective on battery diagnostics.

BALANCING ACCURACY AND ROBUSTNESS

All this complexity presents the BMS with a common challenge: balancing accuracy and robustness.

On one hand, a BMS can try to keep pace with all of the complexity. It may provide accurate results but will struggle with robustness, meaning it can deliver stable and plausible (though not necessarily accurate) SOC estimates but fails when conditions become too complex. Alternatively, a BMS that simplifies its approach will calculate SOC consistently, despite changes in conditions, though the results will not be close to the actual SOC.

Given that the operation of an entire BESS site depends on the BMS SOC, robustness is often prioritized over accuracy.

CLOUD-BASED PREDICTIVE BATTERY ANALYTICS

Unlike BMS, cloud-based predictive battery analytics offer a new perspective on battery diagnostics, serving as an additional layer to enhance the BMS. But, how exactly do cloud-based analytics overcome the issues discussed earlier, especially when they rely on most of the same signals as the BMS for SOC estimation? Predictive battery analytics have three tools in its arsenal that improve SOC accuracy beyond the BMS' capabilities:

1. **Fleet comparisons**
2. **Cloud computational power**
3. **Historical data**

Fleet Comparisons

One of the primary setbacks of a BMS is that it only monitors one battery – the one it is connected to. Even an advanced BMS that is capable of learning from past operations is limited to its own single battery's data. But what if that battery is operated under conditions in which it hasn't been operated before in its entire lifetime, like extreme temperatures or a safety-critical state? To the battery's BMS this is entirely new territory, in which it lacks the data to draw informed conclusions on SOC.

On the other hand, cloud-based predictive battery analytics can tap into data from entire fleets of batteries—not just one—offering unprecedented accuracy is SOC estimation. This approach means predictive analytics can transfer insights gained from one battery to the entire network, enabling comparative analysis that quickly identifies outliers. The power of cloud-based predictive analytics lies in its capacity to aggregate massive amounts of data, making it especially effective in pinpointing and solving systemic issues like SOC inaccuracies caused by BMS.

By benchmarking SOC errors across all deployed batteries, cloud-based analytics can zero in on the exact conditions that are causing the BMS to inaccurately estimate SOC, identify these patterns and help mitigate them to determine an accurate SOC. This method facilitates a more nuanced understanding of SOC estimations, as shown in *Figure 3*, which highlights how BMS SOC calculation errors are benchmarked. The result is a more reliable and precise SOC estimation system that continually improves over time through the analysis of collective battery performance data.

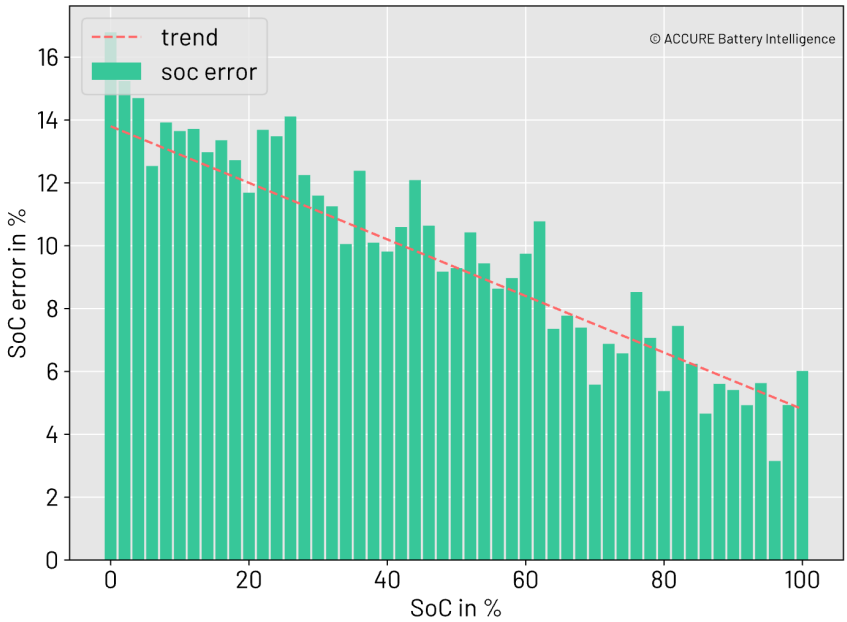


Figure 3

Benchmarking of SOC error across SOC range.

For the example in *Figure 3*, the lowest SOC error is observed at high SOC levels, a common characteristic of BMS algorithms that recalibrate SOC after occasional full charges using the voltage method. The large SOC estimation error at high SOC may be a result of insufficient cell balancing, leading to SOC discrepancies across cells. Additionally, the trend of increasing SOC estimation error with decreasing SOC is typical of BMS algorithms using a capacity value (e.g., nominal capacity) larger than the actual usable capacity.

This is a simple example that demonstrates how cloud-based predictive battery analytics provide valuable insights, sometimes even identifying the root cause of SOC inaccuracies.

Cloud Computing Power

Cloud-based analytics can rely on significantly more computing power than a cost-optimized BMS, allowing for the use of more complex battery models, better handling of complex phenomena like the hysteresis effect, and more accurate translation of voltage measurements to SOC values. *Figure 4* illustrates this in the context of the hysteresis effect discussed in *Figure 2*.

Figure 4 demonstrates how the true OCV (**orange line**) changes over time beginning with a discharge from 60% SOC (**marked by the orange circle**) to 20%. Due to the hysteresis effect, the OCV converges from the charge OCV curve to the discharge OCV curve. When the battery is then charged back to 50%, the OCV starts to converge again, this time toward the charge OCV curve, but it doesn't quite reach it because the battery is discharged to approximately 44% (**marked by the red cross**).

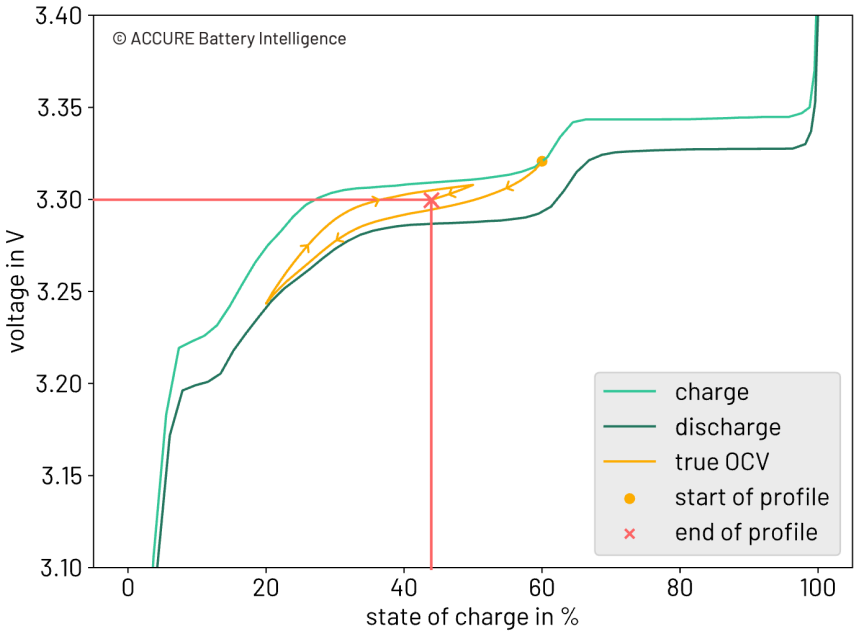


Figure 4

Illustration of the mechanism of converging from one OCV curve to the other using a hysteresis sub-model

The mechanism of converging from one OCV curve to the other can be described by an advanced battery model that includes a hysteresis sub-model, something that is too complex for a BMS, but well within reach for cloud-based predictive analytics. With the ample computational power of cloud-based analytics, a voltage measurement (e.g., 3.3 V) can be translated into a precise SOC value (e.g., 44%). As a reminder: In *Figure 2* (see section on voltage method) for the same voltage measurement we had a notable uncertainty spanning from 27% SOC to 62% SOC using BMS-estimated SOC.

Processing Historic Data

The BMS is short-sighted by default. It operates iteratively, processing a new set of measurements at each time step, and then immediately calculating the next step. While this does mean the BMS is capable of operating in real-time, it limits the BMS' ability to process larger coherent data sets and track long-term trends.

Cloud-based analytics, on the other hand, have access to the entire historical data of a battery, making it easy to monitor long-term trends, and detect and compensate for issues like SOC drift from Ah-counting (as shown in *Figure 5*).

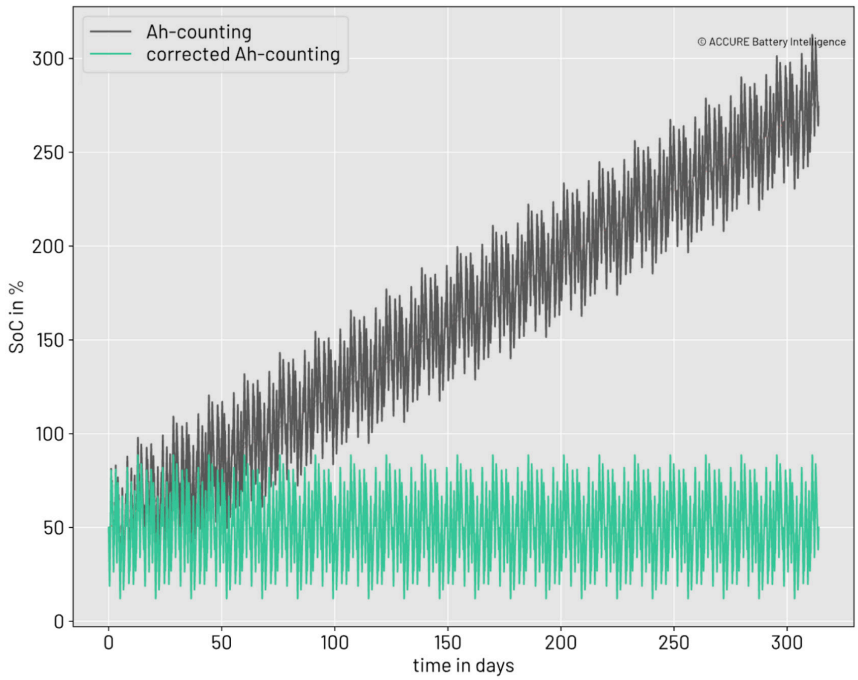


Figure 5

Illustration of an SOC drift that can be compensated for by cloud-based analytics

Cloud-based predictive battery analytics can detect and quantify this trend by estimating the offset current causing the SOC to drift off. Once this trend is identified, cloud-based analytics can compensate for it, resulting in a corrected SOC curve with Ah-counting (light green curve). Instead of reaching a 10% SOC error after 14 days, as shown in *Figure 1*, this method ensures that the cloud-computed SOC remains accurate for weeks and even months without needing frequent recalibration.

These capabilities enable cloud-based predictive battery analytics to achieve accurate SOC estimates within 2% of the actual value, even for challenging battery technologies like LFP batteries under difficult operating conditions.

Integrating the cloud platform with the trading algorithm turns a potential liability into a competitive advantage in the energy market.

USING CLOUD-BASED SOC ESTIMATIONS TO INCREASE REVENUE

Inaccurate SOC estimations in LFP batteries can lead to significant financial consequences, especially when these batteries are used in energy markets and grid services. To capitalize on cloud-generated insights, they must be actively integrated into the operational strategy, which can be done in two ways.

One approach is to manually adjust the trading algorithm using cloud-provided offsets to correct SOC discrepancies. While this approach can improve trading accuracy and reduce financial risk, it is only a partial solution.

But, there is another approach. The true value of analytics is only realized when the trading algorithm is fully integrated with the cloud platform, allowing for continuous, near real-time updates of the SOC. This integration ensures that trading decisions are always based on the most accurate and up-to-date data, enabling operators to:

- **Optimize asset utilization**
- **Minimize penalties for non-compliance with market conditions**
- **Enhance revenue streams**
- **Reduce the need for conservative safety margins in trading**

At its core, integrating the cloud platform with the trading algorithm provides a dynamic and responsive approach to SOC management, turning a potential liability into a competitive advantage in the energy market. Traders can now utilize precise, reliable, and up-to-date data to fully exploit the asset's capabilities, maximizing energy deployment and revenue potential while minimizing financial risks.

Case Study: Increasing Revenue with Accurate SOC Readings

A European energy storage operator was facing major revenue losses due to inaccurate state-of-charge (SOC) estimations from their Battery Management System (BMS). These inaccurate SOC calculations led to lost trading opportunities, penalties, and warranty complications. To address this, the operator turned to ACCURE Battery Intelligence for a solution.

ACCURE analyzed data from the operator's system and applied insights from similar systems around the world. By applying a unique LFP SOC estimation, ACCURE was able to identify discrepancies between the BMS-calculated SOC and the actual SOC, creating a corrective model to improve accuracy.

As a result, ACCURE implemented an early SOC detection system to catch error build-ups, enabling recalibrations during periods of low demand. This approach has reduced SOC errors to within ±3%, a significant improvement from previous deviations of up to 50%.

Cloud-based predictive battery analytics plays a significant role in increasing system profitability by enabling accurate SOC readings. With ACCURE's insights, the operator achieved more reliable trading, minimized penalties, and optimized revenue.

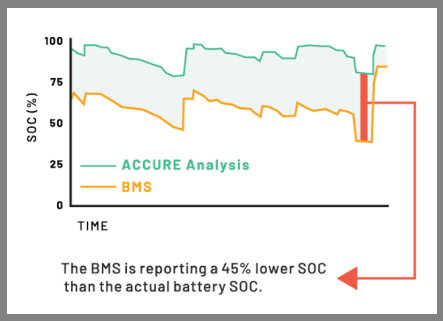


Figure 6

Consistent SOC estimation error by the BMS corrected with cloud-based predictive analytics

CONCLUSION

While LFP batteries offer significant advantages for large-scale energy storage, their inherent challenges with accurate SOC estimation cannot be ignored. Traditional BMS struggle to provide precise SOC readings, often leading to costly inefficiencies and performance issues.

However, by relying on cloud-based predictive battery analytics, operators can significantly improve SOC accuracy, reduce errors, and optimize battery performance. This approach not only addresses the limitations of conventional BMS, but also ensures energy storage systems operate safely, efficiently, and profitably, even in the most challenging conditions.

ACCURE Battery Intelligence

ACCURE helps companies reduce risk, improve performance, and maximize the business value of battery energy storage. Our predictive analytics solution simplifies the complexity of battery data to make batteries safer, more reliable, and more sustainable. By combining cutting-edge artificial intelligence with deep expert knowledge of batteries, we bring a new level of clarity to energy storage. Today, we support 3+ GWh for customers worldwide, helping optimize the performance and safety of their battery systems.

Learn more at accure.net

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